Connected cographs with distance eigenvalues just below -2

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Abstract

Lou and Lin proved that no connected cograph has a distance eigenvalue in the open interval $(-2,-1)$. In a recent survey, Manna, Mandal and Tura ask whether some interval $(-2-\varepsilon,-2)$ is also free of distance eigenvalues of connected cographs. It is not: for every $m \geq 1$ the connected cograph $K_1 \vee (K_1 \cup mK_2)$ on $2m+2$ vertices has a distance eigenvalue in $\left(-2-\frac{1}{2m},-2\right)$. The same graphs were used by Conde, Dratman and Grippo for the adjacency matrix, and the answer also follows in a few lines from their lemma; we record it because the question is listed as open, with a direct proof that is formally verified in Lean 4. The proof was found and formalized by AI agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

All graphs are finite and simple. A *cograph* is a graph with no induced path on four vertices [3]. The *distance matrix* of a connected graph G is $D(G) = (d(u,v))_{u,v}$, where d is the graph distance, and its eigenvalues are the *distance eigenvalues* of G. Since $D(G)$ is a real symmetric matrix, they are real.

Lu, Huang and Lou [6] showed that a connected threshold graph has no distance eigenvalue in $(-2,-1)$, and Lou and Lin [5] extended this to all connected cographs. For threshold graphs, Alazemi, Andelić, Koledin and Stanić [1] found a larger interval below -1 that contains no distance eigenvalue other than -2. Manna, Mandal and Tura [7] observe that this larger interval does not extend to cographs, with the example $K_1 \vee (K_1 \cup 3K_2)$, whose distance eigenvalue ≈ -2.155 lies in it, and they ask [7, Open Problem "Distance eigenvalues of cographs below -2 and sharpness", part (i)]:

> Is there $\varepsilon > 0$ such that $(-2-\varepsilon,-2)$ contains no distance eigenvalue of any connected cograph?

For $m \geq 1$ let $G_m = K_1 \vee (K_1 \cup mK_2)$: a vertex c adjacent to all other vertices, a vertex z adjacent only to c, and m disjoint edges a_ib_i $(1 \leq i \leq m)$ whose ends are also adjacent to c. The example of [7] is G_3.

Theorem 1. *For every $m \geq 1$, the graph G_m is a connected cograph with a distance eigenvalue in the open interval $\left(-2-\frac{1}{2m},-2\right)$. Hence the answer to the question above is no: for every $\varepsilon > 0$ some connected cograph has a distance eigenvalue in $(-2-\varepsilon,-2)$.*

The eigenvalue is the smallest root of the cubic p_m of Section 2; numerically it is -2.383, -2.222, -2.155, -2.049 and -2.005 for $m = 1, 2, 3, 10$ and 100.

Relation to the work of Conde, Dratman and Grippo. The graphs G_m are the graphs $K_1 \vee (mK_2 + K_1)$ of Conde, Dratman and Grippo [2, Lemma 6 and its proof], who showed that each has an adjacency eigenvalue $\lambda_0 \in (0, \frac{1}{2m+1})$, so that positive adjacency eigenvalues of connected trivially perfect graphs accumulate at 0. Their paper does not discuss the distance matrix, but our answer follows from their lemma in a few lines (Remark 1); the survey [7] cites [2]. We record the answer because the question is listed as open, and we give a direct proof for G_m itself, which we have formalized.

Remark 1. A connected cograph G on at least two vertices has diameter at most 2, so $D(G) = 2J - 2I - A(G)$ [7, Proposition "Diameter"]. Let x be an eigenvector of $A(G_m)$ for the eigenvalue λ_0 of [2], and let $G = K_1 \vee (G_m \cup G_m)$, a connected cograph. The vector y that is 0 on the new vertex, x on the first copy and $-x$ on the second satisfies $A(G)y = \lambda_0 y$ and $Jy = 0$, so $D(G)y = (-2 - \lambda_0)y$, and $-2 - \lambda_0 \in (-2 - \frac{1}{2m+1}, -2)$. This remark is not part of the formal verification.

2 Proof of Theorem 1

Lemma 1. *G_m is a connected cograph, and for $u \neq v$ we have $d(u,v) = 1$ if u and v are adjacent and $d(u,v) = 2$ otherwise.*

Proof. Every vertex other than c is adjacent to c, so G_m is connected and any two non-adjacent vertices are at distance 2. Suppose that $v_0v_1v_2v_3$ is an induced path. If $v_1 = c$, then c is adjacent to v_3, which is not allowed; if $v_1 = z$, then $v_0 = v_2 = c$, since c is the only neighbour of z. So v_1 is an end of an edge a_ib_i, and its neighbours v_0, v_2 lie in $\{c, a_i, b_i\} \setminus \{v_1\}$. Neither of them is c, since c would be adjacent to the other one; so $v_0 = v_2$, a contradiction. □

By Lemma 1, the partition of the vertices into $\{c\}$, $\{a_1, b_1, \ldots, a_m, b_m\}$ and $\{z\}$ is equitable for $D(G_m)$: the row sums of $D(G_m)$ over the three parts are $(0, 2m, 1)$ for c, $(1, 4m-3, 2)$ for each a_i and b_i, and $(1, 4m, 0)$ for z. Let

$$p_m(x) = x^3 + 3x^2 - x - 3 - m(4x^2 + 10x + 4),$$

the characteristic polynomial of the quotient matrix $\left(\begin{smallmatrix} 0 & 2m & 1 \\ 1 & 4m-3 & 2 \\ 1 & 4m & 0 \end{smallmatrix}\right)$. Then

$$p_m(-2) = 3 > 0, \qquad p_m\Big(-2 - \frac{1}{2m}\Big) = -\frac{4m^2 + 6m + 1}{8m^3} < 0,$$

so by the intermediate value theorem p_m has a root λ with $-2 - \frac{1}{2m} < \lambda < -2$.

We show that λ is a distance eigenvalue of G_m with an explicit eigenvector. Since $\lambda < -2$, $\lambda^2 - 1 \neq 0$; put

$$\alpha = \frac{2m(\lambda + 2)}{\lambda^2 - 1}, \qquad \gamma = \lambda\alpha - 2m,$$

and let x be the vector with $x_c = \alpha$, $x_z = \gamma$ and $x_v = 1$ for the other $2m$ vertices; $x \neq 0$. By the row sums above,

$$\begin{aligned} (Dx)_c &= 2m + \gamma = \lambda\alpha, \\ (Dx)_z &= \alpha + 4m = \lambda\gamma, \\ (Dx)_{a_i} = (Dx)_{b_i} &= \alpha + (4m - 3) + 2\gamma = \lambda. \end{aligned}$$

The first line is the definition of γ. The second is equivalent to $\alpha(\lambda^2 - 1) = 2m(\lambda + 2)$, the definition of α. The third is equivalent to $\alpha(1 + 2\lambda) = \lambda + 3$, that is, after multiplying by $\lambda^2 - 1$, to $2m(\lambda + 2)(2\lambda + 1) = (\lambda + 3)(\lambda^2 - 1)$, which is $p_m(\lambda) = 0$. Hence $Dx = \lambda x$. Given $\varepsilon > 0$, choosing m with $\frac{1}{2m} < \varepsilon$ proves the last sentence of the theorem. □

3 Formal verification

Theorem 1 is formally verified in Lean 4 (version 4.34.1) with Mathlib [8] (version v4.34.1, commit `d13f23b7`). The files are in version 1.18.0 of the repository [4], directory `lean/Research/CographDist/`.

The statement. The statement file `Challenge.lean` contains only definitions and the statement, no proofs. In ASCII notation (implicit arguments omitted):

```
def HasInducedP4 (G : SimpleGraph V) : Prop :=
  exists f : Fin 4 -> V, Function.Injective f /\
    forall i j : Fin 4, G.Adj (f i) (f j) <-> (i.val + 1 = j.val \/ j.val + 1 = i.val)
def IsCograph (G : SimpleGraph V) : Prop := Not (HasInducedP4 G)
def distMatrix (G : SimpleGraph V) : Matrix V V Real := fun u v => (G.dist u v : Real)
def IsDistEigenvalue [Fintype V] (G : SimpleGraph V) (mu : Real) : Prop :=
  exists x : V -> Real, x != 0 /\ (distMatrix G).mulVec x = mu * x
def OpenProblemI_negative : Prop :=
  forall eps : Real, 0 < eps -> exists (V : Type) (_ : Fintype V) (G : SimpleGraph V),
    G.Connected /\ IsCograph G /\
    exists mu : Real, -2 - eps < mu /\ mu < -2 /\ IsDistEigenvalue G mu
```

(In the file, `mu * x` is the scalar multiple `mu • x`, and `distMatrix` is marked `noncomputable`.) Mathlib's `SimpleGraph.dist` is 0 for vertices in different components; the statement asks for connected graphs. A test file checks that the path on four vertices is not a cograph, that K_2 is a connected cograph, and that 1 is and 2 is not a distance eigenvalue of K_2.

Review and freezing. An independent Claude Code agent, which had not written the file, reviewed the statement and the tests against the definitions and the question of [7] and approved them; its three minor suggestions (a comment and three more tests) were applied, and the file was frozen on 9 November 2026 at 22:56 UTC by recording the SHA-256 hash

`960a24b9de500d2449a73a83b7385653f63054ab52dab3053d287c4e1270e569`

of `Challenge.lean`. The proof files were drafted while the review was running; the statement file was not changed to fit them, and after the review only its comment changed. Moving the files to the directory of the repository changed only their `import` lines; `Challenge.lean` itself, which imports only Mathlib, did not change.

The proof. Three files prove the theorem along the lines of Section 2: `Family.lean` (the graph G_m, its connectedness, its distances and the absence of an induced P_4), `Eigen.lean` (the root of p_m by the intermediate value theorem, and the eigenvector) and `Main.lean` (`openProblemI_negative`, and `family_theorem`, the statement of Theorem 1 for each G_m). The final audit `Audit.lean` restates both and prints their axioms and those of the five tests: each is exactly `propext`, `Classical.choice` and `Quot.sound`. No file uses `sorry`, `native_decide`, `bv_decide` or an `axiom` declaration.

Checks. On 9 November 2026 (23:20–23:23 UTC) the six files were compiled, one module at a time, in two new empty output directories, one after the other, on the author's Mac, with the pinned toolchain and Mathlib, which were not rebuilt; each module was then replayed with `leanchecker`. All 24 commands ended with exit code 0, with no error, no warning and no `sorry`, and the compiled `.olean` files of the two directories are byte for byte identical. From 9 to 10 November 2026 the whole Lean project of the repository, with these six files, was rebuilt from a new empty directory on the author's Windows computer, one module at a time and not with `lake`: all 3005 modules compiled and 3004 passed `leanchecker`; the remaining one, the root of the project, has only imports and no constants. The formalization has not been checked with comparator or with an independent kernel.

How this paper was produced

This work was carried out on 9 November 2026 (UTC) in a workflow directed by the author, using agents of Anthropic Claude Code with the model Claude Opus 5.5. A Claude Code agent scouting for open problems found the question, noticed that the family G_m answers it, and checked this numerically and symbolically. Another Claude Code agent searched the literature (22:49–23:17 UTC) and found the lemma of [2] and Remark 1. Claude Code wrote the statement file, the proofs and this text and ran the checks of Section 3. An independent Claude Code agent (model Claude Fable 5.1), which had written none of the files, reviewed the statement file and compiled it with its tests; before the release a second such agent (same model) reviewed this paper, the Lean files and the records of the builds, without running Lean. The corrections of both reviews were applied. OpenAI Codex was not involved. Throughout, "we" refers to this workflow; the reading and searching described in the paper were done by the agents. The author is not a professional mathematician. The author has read this paper and, using a side-by-side table, compared its Lean statement with Theorem 1. No human mathematician has reviewed this paper. The correctness of the result rests on the Lean proof and on the agreement of the Lean statement with the question of [7], which was checked by the reviewing agent; the written proof is given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

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